

SEC EDGAR

Financial KPI Impact Analysis

COVID-19 Disruption & Recovery Study

DATA COVERAGE Q1 2018 — Q4 2024	COMPANIES ANALYZED 139,696 Filings	KPIs TRACKED 8 Financial Ratios
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Data Source: SEC EDGAR DERA Financial Statement Data Sets	Platform: Databricks Delta Lake SQL	

Executive Summary

Core Research Question

Which financial KPIs derived from SEC EDGAR filings were most impacted by COVID-19, how did they recover Post-COVID, and do these patterns vary by industry sector?

This report analyzes 139,696 SEC EDGAR financial filings spanning Q1 2018 through Q4 2024, examining the impact of COVID-19 on eight key financial performance indicators (KPIs) across publicly traded companies. The analysis is structured across three distinct periods: Pre-COVID (2018-2019), COVID-Impact (Q1-Q2 2020), and Post-COVID Recovery (Q3 2020-2024).

Key Findings at a Glance

Finding	KPI	Direction	Magnitude
Liquidity improved strongly Post-COVID	Current Ratio	▲ Improved	2.88 → 3.47
Leverage steadily declined	Debt Ratio	▼ Declined	0.82 → 0.69
Profitability recovering slowly	ROA	▲ Slow Recovery	-0.13 → -0.10
Operating efficiency not recovered	Operating Margin	▼ Deteriorated	-0.17 → -0.22
Debt burden increased during COVID	Debt to Equity	▲ Increased	0.43 → 0.61
37.7% of companies improved	ROA (Company Level)	▲ Improving	1,743 companies
Eating & drinking worst hit sector	ROA (SIC 5810)	▼ -0.54 delta	COVID vs Pre-COVID
Retail fastest recovery	ROA (SIC 5900)	▲ +0.52 delta	Post vs Pre-COVID

1. Methodology & Data Overview

1.1 Data Source

The analysis uses SEC EDGAR DERA Financial Statement Data Sets — structured quarterly releases of XBRL-tagged financial data from all public company filings. Data was ingested, cleaned and processed on Databricks using Apache Spark and Delta Lake.

1.2 Period Classification

Period	Date Range	Quarters	Filings
Pre-COVID	Q1 2018 — Q4 2019	8	~5,800 avg/quarter
COVID-Impact	Q1 2020 — Q2 2020	2	~4,777 avg/quarter
Post-COVID Recovery	Q3 2020 — Q4 2024	18	~7,670 avg/quarter

1.3 KPIs Analyzed

KPI	Formula	Measures
Current Ratio	Current Assets / Current Liabilities	Short-term liquidity
Debt Ratio	Total Liabilities / Total Assets	Overall leverage
Return on Assets (ROA)	Net Income / Total Assets	Asset efficiency
Return on Equity (ROE)	Net Income / Stockholders Equity	Equity efficiency
Operating Margin	Operating Income / Revenue	Operational efficiency
Gross Margin	Gross Profit / Revenue	Production efficiency
Asset Turnover	Revenue / Total Assets	Asset utilization
Debt to Equity	Long Term Debt / Stockholders Equity	Financial leverage

1.4 Data Quality Notes

- Segment-level financial data was excluded — only consolidated company-level figures retained
- Outlier filtering applied: Current Ratio 0-50, Debt Ratio 0-10, ROA -5 to 5, ROE -5 to 5
- 1,254 companies with missing SIC codes retained but excluded from sector-level analysis
- Debt Ratio showed extreme Pre-COVID outliers — median values used for sector comparisons

2. Macro KPI Trend Analysis

The following analysis presents market-wide KPI averages across all three periods, providing a macro-level view of how COVID-19 impacted corporate financial health across the US public equity universe.

2.1 KPI Summary by Period

KPI	Pre-COVID	COVID-Impact	Post-COVID	Recovery Status
Current Ratio	2.8816	2.9175	3.4700	✅ Exceeded baseline
Debt Ratio	0.8231	0.7599	0.6904	✅ Fully recovered
ROA	-0.1293	-0.1104	-0.0959	⚠️ Recovering slowly
ROE	0.0001	0.0205	-0.0579	❌ Below baseline
Operating Margin	-0.1676	-0.1576	-0.2204	❌ Not recovered
Gross Margin	0.4126	0.4184	0.4070	⚠️ Slight decline
Asset Turnover	0.8765	0.8291	0.8087	❌ Declining
Debt to Equity	0.4292	0.6115	0.5283	⚠️ Partially recovered

2.2 Liquidity Analysis — Current Ratio

The current ratio demonstrated the strongest and most consistent recovery trajectory of all KPIs analyzed. From a Pre-COVID average of 2.88, it remained relatively stable during COVID-Impact (2.92) before surging to 3.47 Post-COVID — exceeding the Pre-COVID baseline by 20.4%. This reflects aggressive cash reserve building by companies in the aftermath of the pandemic, driven by lessons learned during the liquidity crisis of early 2020.

2.3 Leverage Analysis — Debt Ratio

The debt ratio showed a consistent and sustained decline across all three periods (0.82 → 0.76 → 0.69), indicating that companies systematically reduced their leverage over the analysis window. While debt-to-equity spiked during COVID-Impact (0.43 to 0.61) as companies drew on credit facilities and issued debt to survive, the overall balance sheet deleveraging trend is a positive structural signal for financial stability.

2.4 Profitability Analysis — ROA & Operating Margin

Profitability metrics present a more nuanced picture. ROA improved gradually from -0.13 Pre-COVID to -0.10 Post-COVID, indicating slow but directionally positive recovery. However, operating margin deteriorated from -0.17 Pre-COVID to -0.22 Post-COVID — suggesting that while companies are generating marginally more profit relative to assets, their core operations remain under structural cost pressure. This divergence warrants close monitoring.

3. Sector-Level Impact Analysis

Sector-level analysis reveals significant dispersion in COVID impact and recovery trajectories. The following analysis focuses on sectors with 10 or more companies for statistical reliability.

3.1 Most Impacted Sectors During COVID

SIC	Industry	Pre-COVID ROA	COVID ROA	ROA Delta	Op Margin Delta
5810	Eating & Drinking Places	-0.1870	-0.7266	-0.5397	-0.0916
8741	Management Consulting	0.4296	-0.0647	-0.4942	0.0001
2030	Food Processing	0.2892	-0.0566	-0.3458	-0.0169
6794	Patent Owners & Lessors	-0.0726	-0.3981	-0.3255	-0.4683
3829	Measuring Instruments	-0.0564	-0.3602	-0.3039	-0.0769
4899	Communications Services	0.0382	-0.2592	-0.2973	-0.0519
3670	Electronic Components	-0.1618	-0.4360	-0.2743	0.0491

The eating and drinking sector (SIC 5810) experienced the most severe COVID impact with an ROA decline of -0.54, directly attributable to mandatory closures, capacity restrictions, and consumer behavioral changes during the pandemic. Management consulting and food processing sectors also suffered significant profitability disruptions.

3.2 Fastest Recovering Sectors Post-COVID

SIC	Industry	Pre-COVID ROA	Post-COVID ROA	ROA Recovery
5900	Retail Stores	-0.5669	-0.0512	+0.5157
8082	Home Health Services	-0.4693	-0.0250	+0.4443
3826	Laboratory Instruments	-0.5215	-0.0876	+0.4338
4700	Transportation Services	-0.4509	-0.0553	+0.3956
3842	Orthopedic Equipment	-0.4823	-0.1193	+0.3630
7380	Miscellaneous Business Services	-0.5125	-0.1642	+0.3483
4011	Railroads	-0.2292	0.0830	+0.3122

Retail stores demonstrated the strongest Post-COVID recovery, benefiting from the reopening of physical retail, pent-up consumer demand, and e-commerce acceleration. Healthcare-adjacent sectors including home health services and orthopedic equipment also recovered strongly, driven by the post-pandemic healthcare spending surge. Transportation services recovery reflects the normalization of supply chains and travel.

4. Leverage & Debt Trajectory Analysis

The leverage analysis tracks debt ratio and debt-to-equity on a quarterly basis across all 28 quarters, providing a granular view of how corporate balance sheets evolved through the COVID cycle.

4.1 Quarterly Leverage Trends

Period	Year	Q	Avg Debt Ratio	Avg D/E	Avg Current Ratio
Pre-COVID	2018	1	0.7382	0.4877	2.8576
Pre-COVID	2019	4	0.8201	0.6032	2.9186
COVID-Impact	2020	1	0.7002	0.7031	2.9678
COVID-Impact	2020	2	0.8088	0.5347	2.8748
Post-COVID	2021	1	0.6598	0.4329	3.4549
Post-COVID	2022	1	0.5875	0.6624	4.1366
Post-COVID	2023	4	0.7322	0.5731	3.0975
Post-COVID	2024	4	0.7429	0.7629	2.9582

4.2 Key Leverage Observations

- Debt-to-equity spiked to 0.70 in Q1 2020 — companies drew on credit facilities immediately as COVID hit
- Debt ratio reached its lowest point in Q1 2022 (0.59) — peak balance sheet health Post-COVID
- Current ratio peaked at 4.14 in Q1 2022 — companies holding maximum cash reserves
- Both debt ratio and D/E rising again in 2024 (0.74, 0.76-0.79) — new leverage cycle beginning
- Normalization of current ratio in 2024 (2.9-3.1) suggests companies deploying excess cash

Analyst Alert — 2024 Leverage Uptick

The simultaneous increase in debt ratio and debt-to-equity in 2024, combined with declining current ratios, suggests companies are entering a new leverage cycle — potentially driven by M&A activity, capital expenditure programs, or refinancing of COVID-era debt. Asset managers should monitor leverage trends closely heading into 2025.

5. Company-Level Recovery Classification

Individual company analysis classifies all companies present in both Pre-COVID and Post-COVID periods based on their ROA trajectory, providing a bottom-up view of financial recovery.

5.1 Recovery Classification Results

Classification	Companies	% of Total	Avg ROA Change	Avg Op Margin Change
Improving	1,743	37.7%	+0.2118	+0.1501
Deteriorating	1,651	35.7%	-0.1598	-0.0865
Stable	1,327	28.6%	0.0000	+0.0133

5.2 Interpretation

The distribution of improving vs deteriorating companies is notably balanced — 37.7% improving against 35.7% deteriorating — suggesting COVID created significant bifurcation in corporate financial performance rather than a uniform market-wide recovery. The magnitude of ROA improvement among recovering companies (+0.21) exceeds the deterioration magnitude (-0.16), indicating that companies that benefited from COVID tailwinds improved more dramatically than those that suffered.

- Improving companies gained on average +0.21 ROA and +0.15 operating margin — meaningful structural improvement
- Deteriorating companies lost -0.16 ROA and -0.09 operating margin — less severe than the gains of improvers
- Stable companies showed marginal operating margin improvement (+0.013) suggesting broad operational efficiency gains across the market
- The 37.7% vs 35.7% split suggests COVID was a net neutral event at the company level, but with significant winners and losers

6. KPI Volatility Analysis

Standard deviation of KPI values across periods measures the consistency of financial performance signals — higher volatility indicates less reliable and predictable metrics.

6.1 KPI Volatility by Period

KPI	Pre-COVID StdDev	COVID StdDev	Post-COVID StdDev	Most Volatile Period
Current Ratio	4.2401	4.2914	5.1712	Post-COVID
Debt Ratio	1.1542	1.0556	0.9848	Pre-COVID
ROA	0.5710	0.4729	0.4362	Pre-COVID
ROE	0.8556	0.8392	0.8391	Pre-COVID
Operating Margin	0.7930	0.8074	0.8368	Post-COVID
Gross Margin	0.2398	0.2306	0.2471	Post-COVID
Asset Turnover	0.8840	0.7860	0.8313	Pre-COVID
Debt to Equity	1.4757	1.3432	1.4759	Pre-COVID / Post-COVID

6.2 Volatility Insights

- Current ratio became most volatile Post-COVID (5.17) — divergence between cash-rich and cash-poor companies widening
- Operating margin volatility increased Post-COVID (0.84) — structural cost pressures creating performance dispersion
- ROA volatility actually declined Post-COVID (0.44) — companies converging toward similar asset efficiency levels
- Debt to equity remains consistently volatile across all periods (~1.4-1.5) — leverage strategies vary widely by company

7. Conclusions & Strategic Implications

7.1 Core Findings

Finding 1 — Liquidity is the clearest COVID success story

Current ratios improved across all periods and ended Post-COVID significantly above the Pre-COVID baseline. Companies that survived COVID emerged with stronger liquidity positions, reflecting both forced cash conservation during the pandemic and sustained conservative treasury management thereafter.

Finding 2 — Operating efficiency is the most concerning laggard

Operating margin is the only major KPI that deteriorated through the full period and has not recovered to Pre-COVID levels. This suggests structural cost inflation — likely driven by labor cost increases, supply chain normalization costs, and inflationary pressures — is compressing margins across the corporate landscape. This is a key risk for equity investors.

Finding 3 — Leverage is entering a new cycle

After a deleveraging period from 2020-2022, both debt ratio and debt-to-equity are rising again in 2024. Combined with declining current ratios, this suggests companies are deploying their COVID-era cash reserves into growth investments or shareholder returns while simultaneously taking on new debt. This leverage re-expansion warrants monitoring.

Finding 4 — COVID created significant corporate bifurcation

The 37.7% improving vs 35.7% deteriorating company split indicates COVID was not a uniform shock — it created significant winners and losers. Sectors aligned with digitalization, healthcare, and essential services improved, while contact-intensive and cyclical sectors deteriorated.

7.2 Implications by Audience

Audience	Key Implication	Action
Asset Managers	Operating margin compression is systemic — not company-specific	Screen for companies with improving operating margins in sectors showing recovery
Trading Firms	Leverage re-expansion in 2024 signals potential credit risk	Monitor debt covenant compliance and refinancing risk for 2024-2025
REITs	Current ratio improvement suggests stronger tenant balance sheets	REIT landlords may benefit from improved tenant creditworthiness
Credit Analysts	35.7% of companies still deteriorating Post-COVID	Enhanced credit monitoring for companies in deteriorating SIC sectors
Risk Analysts	KPI volatility increasing Post-COVID in current ratio and operating margin	Use volatility-adjusted ratio thresholds rather than static benchmarks

7.3 Data Limitations

- Analysis covers US-listed public companies only — private company trends may differ significantly
- SIC codes are self-reported and may not accurately reflect business model evolution
- Outlier filtering removed ~7% of observations — extreme cases excluded from trend analysis
- Stock price data not incorporated — ratio-to-return correlations remain for future analysis
- COVID-Impact period limited to 2 quarters — may not fully capture the initial shock duration

Appendix — Technical Reference

A. Pipeline Architecture

Layer	Tool	Output
Ingestion	Python + Requests + Databricks	56 raw Delta tables (28 quarters × 2 files)
Master Tables	PySpark unionByName	edgar_sub_all, edgar_num_all
Cleaning	PySpark + Spark SQL	edgar_sub_clean, edgar_num_clean
Pivot	Spark SQL CASE WHEN MAX	edgar_ratios (191,828 rows)
Ratio Computation	Spark SQL	8 individual ratio tables
KPI Unified	Spark SQL LEFT JOIN	edgar_kpi_unified (150,347 rows)
Outlier Filtering	Spark SQL BETWEEN	edgar_kpi_clean (139,696 rows)
Analysis	Databricks SQL	7 analytical queries

B. KPI Definitions & Outlier Bounds

KPI	Formula	Min Bound	Max Bound	Coverage
Current Ratio	Current Assets / Current Liabilities	0	50	150,347
Debt Ratio	Total Liabilities / Total Assets	0	10	160,891
ROA	Net Income / Total Assets	-5	5	52,401
ROE	Net Income / Stockholders Equity	-5	5	49,531
Operating Margin	Operating Income / Revenue	-5	5	29,082
Gross Margin	Gross Profit / Revenue	-2	2	16,793
Asset Turnover	Revenue / Total Assets	0	10	35,181
Debt to Equity	Long Term Debt / Stockholders Equity	-10	10	9,662

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